

DAV Quiz 2 - B

Roll No: _____

Name: _____

Instructions: Do not write on the question paper or tick any option. Write selected option inside the brackets given at the end of each question "Answer []"

1. What role does the transaction log play in ensuring reader see a "single source of truth"?

A) It stores copies of the entire table dataset directly in memory. B) It dynamically rewrites immutable physical data files in-place during active scans. C) It acts as a database gatekeeper, blocking non-SQL query engines from accessing data. D) It tracks exactly which physical Parquet files are active, shielding readers from partial writes, failed jobs, or orphaned files. Answer []

2. How does Delta Lake execute a logical DELETE or UPDATE operation on immutable Parquet files?

A) It performs a "Copy-on-Write," creating a new Parquet file with the changes and recording a "soft-delete" (tombstone) in the log. B) It accesses the physical disk blocks and overwrites the exact bits containing those rows in-place. C) It converts the Delta table into a highly normalized Data Vault format. D) It forces the compute cluster to reboot and clears the memory cache. Answer []

3. What is the primary functional difference between the Silver and Gold layers in a Medallion Architecture?

A) Silver contains raw data; Gold contains slightly compressed raw files. B) Silver standardizes and cleanses source-oriented data; Gold curates, integrates, and optimizes data specifically for business consumption. C) Silver is optimized for vertical scaling; Gold is designed to run only on on-premises relational databases. D) Silver uses schema-on-read; Gold is completely schema-less. Answer []

4. Why is the Gold layer often referred to as the "pinnacle of data refinement"?

A) Because it strictly houses raw, unjoined system duplicates. B) It decouples physical file networks to prevent users from querying with SQL. C) It is optimized for high-performance querying and aggregates, transforming complex data structures into actionable insights. D) It runs continuously in-memory without persisting any physical data to cloud storage. Answer []

5. Why do some data platforms utilize a "One Big Table" (OBT) denormalized layout in the Gold layer instead of a Star Schema?

A) To simplify data consumption for non-SQL users and maximize cloud infrastructure query performance by avoiding expensive joins. B) It reduces the storage footprint by highly normalizing relational records. C)

Because columnar databases cannot physically support tables with foreign keys. D) To force analytical consumers to write complex, nested join logic. Answer []

6. Why is "idempotency" considered a fundamental design principle for Medallion data pipelines?

A) It ensures that running a pipeline multiple times with the same input data always produces the exact same output state without duplicates. B) It allows schemas to evolve blindly without tracking changes in the transaction log. C) It forces the system to run in a single-tier coupled environment. D) It requires data engineers to manually clean up files after every pipeline failure. Answer []

7. What is the difference between schema enforcement and schema evolution in Delta Lake?

A) Enforcement is schema-on-read; evolution is schema-on-write. B) Enforcement blocks writes with mismatched structures to prevent pollution; evolution allows the table schema to safely adapt to new columns. C) Enforcement requires highly normalized databases; evolution is only compatible with unstructured text files. D) There is no difference; both are terms used to describe manual table schema reloads. Answer []

8. Which command can physically restrict long-term "time travel" capabilities on a Delta table?

A) OPTIMIZE B) RESTORE C) VACUUM D) SHALLOW CLONE Answer []

9. In the context of Medallion Architecture, where should schema drift primarily be handled?

A. During ingestion into the Bronze layer, by dropping any non-conforming records before landing. B. During the transition from the Bronze layer to the Silver layer, through explicit schema evolution, parsing, or validation rules. C. In the Gold layer, by re-aggregating tables whenever upstream sources change. D. At the source database level, by enforcing database triggers that block schema changes. Answer []

10. In an enterprise data pipeline, an engineer chooses an Incremental Load using a Delta MERGE operation over a Full Load (truncate-and-replace) for updating a Silver-layer sales table. What is the primary operational trade-off of this decision?

A. Higher storage consumption, because Incremental Loads duplicate historical data on every run. B. Increased pipeline state complexity, as the pipeline must track state/watermarks and handle record updates/upserts. C. Inability to process schema drift, because Incremental Loads require fixed schemas while Full Loads do not. D. Slower execution times for daily batches compared to processing the entire multi-year history. Answer []

Answer

1. D
2. A
3. B
4. C
5. A
6. A
7. B
8. C
9. B
10. B